

## RESEARCH ARTICLE

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# The importance of scale in understanding the effect of stream intermittence on aquatic macroinvertebrate diversity

David E. Wooster  | Sandra J. DeBano

Hermiston Agricultural Research & Extension Center, Oregon State University, Hermiston, Oregon

## Correspondence

David E. Wooster, Hermiston Agricultural Research & Extension Center, Oregon State University, 2121 S 1st St, Hermiston, OR 97838.

Email: david.wooster@oregonstate.edu

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## Abstract

Biotic richness in intermittent reaches ( $\alpha$ -diversity) is generally lower than in perennial reaches. However, variability among intermittent reaches ( $\beta$ -diversity) is predicted to be higher. Yet how patterns in  $\alpha$ - and  $\beta$ -diversity scale up to influence stream network diversity is unclear and has received little attention. In addition, the timing of sampling relative to the resumption of flow in intermittent reaches should influence patterns in diversity. We predicted that early in the wet season, assemblages in intermittent reaches would be recovering from drying and have lower  $\alpha$ -diversity and higher  $\beta$ -diversity relative to perennial reaches but that these patterns might not be reflected at the stream network scale. In contrast, we predicted that late in the wet season, assemblages in intermittent reaches would recover and  $\alpha$ ,  $\beta$ , and network diversity would be similar between intermittent and perennial reaches. These predictions were tested with macroinvertebrate assemblages in an eastern Oregon stream network. Early in the wet season,  $\alpha$ -diversity and network diversity were lower in intermittent than perennial reaches; however,  $\beta$ -diversity was similar between the two types of reaches. Late in the wet season, both  $\alpha$ -diversity and  $\beta$ -diversity were similar between intermittent and perennial reaches, suggesting recovery of intermittent reaches from drying. However, at the network scale, intermittent reaches continued to have lower diversity indicating a lack of recovery. Rare taxa were particularly vulnerable to flow intermittency. Our results indicate that a consideration of spatial scale, taxa abundance/rarity, and timing of sampling will improve our understanding of the effect of flow intermittence on diversity.

## KEYWORDS

diversity, intermittent stream, macroinvertebrates, spatial scale

## 1 | INTRODUCTION

Low-order headwater streams are an important component of drainage networks, making up the majority of the linear extent of channels (Benda, Hassan, Church, & May, 2005). They also provide key ecosystem functions in drainage networks as they are the primary sources of water and sediment and play roles in organic matter processing and nutrient cycling (Bernhardt et al., 2005; Clarke, Mac Nally, Bond, & Lake, 2008; Wallace, Eggert, Meyer, & Webster, 1997). Many headwater streams (or stream reaches) are seasonally intermittent in which

surface water ceases to flow in a predictable fashion during the dry season (Leigh et al., 2016). Growing demands on surface water and climate change are expected to increase the spatial and temporal extent of intermittent reaches, and current and future management of watersheds will require a greater understanding of the dynamics of headwater systems and intermittent reaches (Larned, Datry, Arscott, & Tockner, 2010).

Intermittent flow has strong effects on stream ecosystems. For example, important ecosystem functions such as the rate of leaf litter breakdown can significantly decrease in intermittent reaches

compared with those that run year-round, with consequences for nutrient cycling and food web interactions (Langhans & Tockner, 2006). In addition, intermittence can affect the diversity and structure of biotic assemblages (Stubbington, England, Wood, & Sefton, 2017). For example, intermittent reaches often have low  $\alpha$ -diversity relative to nearby permanent reaches (Soria, Leigh, Datry, Bini, & Bonada, 2017; Stubbington et al., 2017). In contrast, intermittent reaches are expected to have relatively high  $\beta$ -diversity in which assemblages differ among reaches to a greater extent than assemblages do among permanent reaches (Larned et al., 2010). Mechanisms for relatively high  $\beta$ -diversity in intermittent reaches include decreased hydrological connectivity, which limits dispersal and mixing of aquatic taxa among sites (Larned et al., 2010; Rolls, Heino, & Chessman, 2016) and greater environmental heterogeneity among intermittent reaches relative to permanent reaches (Schriever & Lytle, 2016). In addition, although assemblages in taxa-poor intermittent reaches might include specialist taxa not found in perennial reaches (Bogan, Boersma, & Lytle, 2013; Dieterich & Anderson, 2000), many are often simply nested subsets of the assemblages found in taxa-rich perennial reaches (Arscott, Larned, Scarsbrook, & Lambert, 2010; Datry, 2012; Datry et al., 2014). Nestedness indicates that intermittency acts as a filter on the regional taxa pool and regional richness is simply lost in intermittent reaches.

Although there is a burgeoning literature on  $\alpha$ - and  $\beta$ -diversity of intermittent streams, little work has examined how patterns in diversity at these levels scale up to patterns in drainage network or regional diversity ( $\gamma$ -diversity). Given that many studies have found that intermittent reaches have lower  $\alpha$ -diversity than perennial reaches, one might expect intermittence to decrease network diversity. However, this effect might be offset by high  $\beta$ -diversity in intermittent reaches. In one of the few studies examining  $\alpha$ ,  $\beta$ , and network diversity in intermittent and perennial reaches, Storey, Parkyn, Neale, Wilding, and Croker (2011) found that intermittent and permanent reaches had similar  $\alpha$ -diversities but intermittent reaches had higher  $\beta$ -diversities and higher network diversity. The authors attributed higher  $\beta$  and network diversity to greater habitat diversity in intermittent reaches. However, it is not clear whether these relationships between  $\alpha$ ,  $\beta$ , and network diversity result in a general pattern or vary with respect to stream system.

In addition, the role of seasonal timing of flow resumption in influencing these patterns of diversity in intermittent streams is not well understood. Intermittent reaches are highly dynamic systems cycling through wet and dry phases. As the wet phase commences, aquatic fauna can establish in intermittent reaches from resistant stages that have remained in the dry channel and/or through colonization from nearby flow refuges (Leigh et al., 2016; Robson, Chester, & Austin, 2011). During this establishment period, taxa richness increases in intermittent reaches. For example, taxa richness in intermittent headwater streams in Costa Rica increased through the wet season, becoming similar to the richness in nearby perennial streams in ~5 months (de Szoeko, Crisman, & Thurman, 2016). In addition, as the establishment period progresses,  $\beta$ -diversity among intermittent reaches should decline and approach that of perennial reaches as taxa mix among sites (Datry, Moya, Zubieta, & Oberdorff, 2016). Therefore, patterns in diversity among intermittent and perennial reaches and

across scales are dependent upon the timing of sampling relative to the commencement of surface water flow in intermittent reaches.

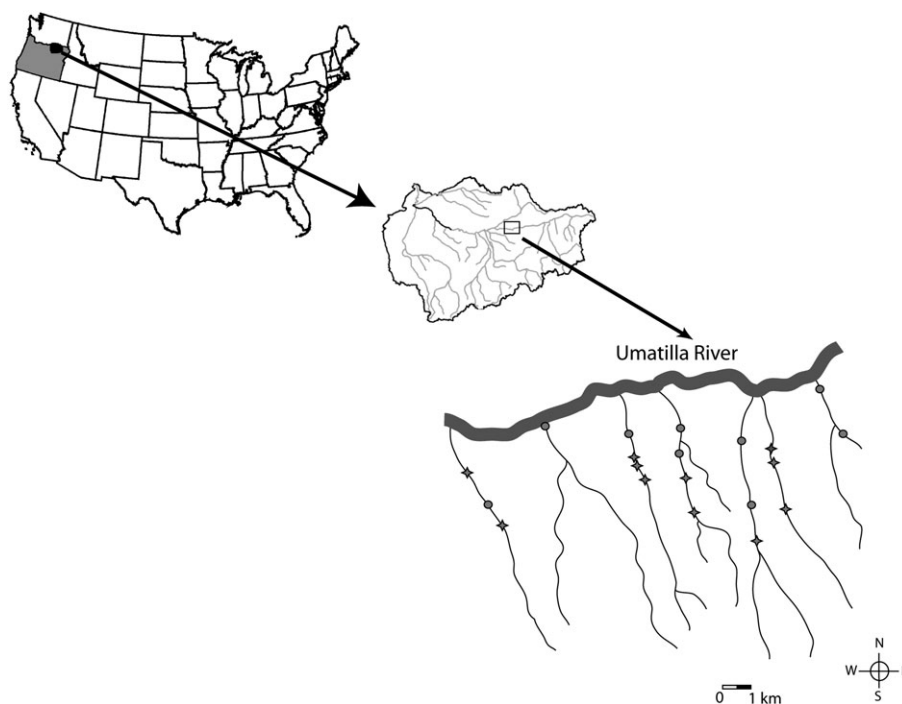
The taxa present in a stream system can also affect the spatial and temporal patterns in diversity in intermittent streams. Taxa-specific qualities can influence the rate of recovery in intermittent reaches. Some taxonomic groups appear particularly sensitive to intermittence (e.g., Ephemeroptera–Plecoptera–Trichoptera [EPT] taxa; Arscott et al., 2010), whereas others are insensitive (e.g., Diptera; Datry et al., 2014). The relative sensitivity of different taxonomic groups might be driven by specific traits the groups possess. Traits conferring either resistance to drying (e.g., desiccation resistant eggs or pupae) or resilience (e.g., rapid dispersal abilities) increase the rate of recovery of assemblages (Robson et al., 2011; Vander Vorste, Corti, Sagoui, & Datry, 2016). Assemblages dominated by relatively insensitive taxa or taxa with traits conferring resistance or resilience will recover quickly resulting in similarities in diversity between intermittent and perennial reaches. Another consideration is the distribution and abundance of specific taxa. The ability of taxa to recolonize intermittent reaches is positively influenced by their abundance in surrounding flow refuges and permanent reaches simply because most taxa produce a large number of propagules (Gotelli & Taylor, 1999). Therefore, systems with a high number of rare taxa will likely take longer to recover than systems with relatively few.

Here, we report the results of a study designed to examine the diversity (taxa richness) of macroinvertebrates in intermittent and permanent reaches of headwater streams in north-eastern Oregon. Macroinvertebrates were sampled in the winter, 5–6 weeks after the resumption of flow in intermittent reaches, and in the summer, after intermittent reaches had been flowing for ~6 months. Taxa richness was examined at three scales, the reach scale ( $\alpha$ -diversity), as variability among reaches ( $\beta$ -diversity), and the stream network scale. Our specific objective was to examine changes in the relationship between these three levels of diversity soon after the commencement of flow (winter) and after 6 months of recovery (summer). We predicted that in winter,  $\alpha$ -diversity would be lower and  $\beta$ -diversity higher in intermittent reaches relative to perennial reaches. However, it was not clear how this pattern would scale up to the network level. In contrast, in summer, we predicted that the assemblages in intermittent reaches would recover to that of perennial reaches and all three levels of diversity would be similar between intermittent and perennial reaches. We examined these patterns for the full assemblages as well as for particular groups predicted to be either particularly sensitive (EPT taxa, poor resistance, poor resilience, and rare) or insensitive (Diptera) to intermittence.

## 2 | METHODS

### 2.1 | Study area

Macroinvertebrate assemblages were collected from 20 reaches spread across seven first- and second-order streams draining the Blue Mountain foothills in the Umatilla River basin in north-eastern Oregon (Figure 1). The Umatilla River basin is a ~5,900-km<sup>2</sup> watershed with a land use dominated by agriculture. The study area has a “semi-arid



**FIGURE 1** The study area and sites. The upper panel shows the location of the Umatilla River basin in north-eastern Oregon. The middle panel shows the Umatilla River basin with a rectangle outlining the study area. The bottom panel is a drawing of the seven first- and second-order study streams that flow into the Umatilla River. Diamonds indicate intermittent sites, and circles indicate perennial sites

cool" climate (Koppen classification: BSk) with warm dry summers and cold, wet winters. Dryland wheat production is the dominant land use and wheat fields surrounded all study sites. All seven streams originate from springs and seeps in the Blue Mountains at an elevation of ~1,100 m and all drain in a northerly direction directly into the fifth-order Umatilla River at an elevation of ~420 m. The streams drain Columbia River basalt and are bordered by alluvium composed of basaltic gravel (Hogenson, 1964). The streams and study sites were all in close proximity with an average distance between study sites of 2.9 km and a maximum distance of 7.9 km. Of the 20 study sites, nine were permanent and 11 were intermittent. Intermittent sites dry annually in early to mid-July and remain dry until enough winter precipitation raises the water table (generally January to February; Contor, Hoverson, Kissner, & Volkman, 1996; Wooster personal observation).

## 2.2 | Field sampling

Macroinvertebrates were collected from the study reaches in late-June to early July 2011 ("summer" sampling). Intermittent reaches went dry within 7–20 days after the summer sampling. Starting in October 2011, intermittent reaches were checked every 7–10 days to determine when flow resumed. Intermittent reaches did not start flowing until late January 2012, and macroinvertebrates were sampled at all reaches in the first week of March 2012 ("winter" sampling), 5–6 weeks after the commencement of flow in intermittent reaches.

Study reaches were 100 m long. Macroinvertebrates were sampled at each reach through six kick net samples (each 0.09 m<sup>2</sup> in area) taken from randomly selected locations within riffle habitat. Kick net samples were taken by first scrubbing all cobbles larger than ~7-

cm diameter with a brush and then using a foot to agitate the sampling area for 30 s. The six kick net samples were combined into a single sample to characterize the macroinvertebrate assemblage at each reach.

Macroinvertebrate samples were preserved in 95% ethanol and processed at the BLM/Utah State University National Aquatic Monitoring Center in Logan, Utah. Subsamples from each sample were processed until at least 500 individual macroinvertebrates were counted and identified. Most insect taxa were identified to genus (except for Chironomidae, which were identified to subfamily), and noninsects were identified to family or order (the dominant noninsect groups were Clitellata and Trombidiformes).

## 2.3 | Analyses

We focused on taxa richness as a metric of biodiversity. Total richness and the richness of specific groups (EPT, Diptera, and rare taxa) were examined. Rare taxa were defined as those found at only one or two reaches. In addition, the richness of groups with traits hypothesized to make taxa particularly vulnerable to intermittency was also analysed. These traits conferred either poor resilience (e.g., weak drift propensity and no flying stage) or poor resistance (e.g., lacking traits conferring desiccation tolerance) to taxa. Taxa were assigned to trait categories using the USEPA trait database (at [www.epa.gov/risk/freshwater-biological-traits-database-traits](http://www.epa.gov/risk/freshwater-biological-traits-database-traits); accessed 1/10/17).

## 2.4 | $\alpha$ -Diversity

For reach-level richness, each sampling site was treated as a replicate of either permanent or intermittent reaches. Comparisons between

permanent and intermittent taxa richness were made using a Welch's *t* test. Taxa richness data were examined to determine whether they approximated a normal distribution using both Shapiro–Wilk tests and plotting studentized residuals against a normal distribution. Data were log transformed when necessary.

## 2.5 | $\beta$ -Diversity

We calculated “variation”  $\beta$ -diversity, which is a measure of the variation in community structure within groups (Anderson et al., 2011).  $\beta$ -Diversity was defined as the distance of each site to group centroids, based upon principle coordinate analysis and using Jaccard dissimilarity, with intermittent and perennial reaches as the groups. Using the *betadispers* function in the R package *betapart*, we tested the homogeneity of multivariate dispersion between groups (Baselga & Orme, 2012; Baselga, Orme, Villegger, De Bortoli, & Leprieur, 2018). A significant difference would indicate that one group of sites is more variable (i.e., higher  $\beta$ -diversity) than the other group. Winter and summer data were analysed separately.

We also examined patterns of nestedness for the winter sampling. During this period, all intermittent reaches had lower taxa richness than any of the permanent reaches (intermittent taxa richness range: 10–23; permanent taxa richness range: 24–39). As a result of this pattern in richness, nestedness analysis allowed a determination of whether the assemblages in intermittent reaches were nested subsets of those found in perennial reaches. We tested for nestedness with presence–absence data and the Brualdi and Sanderson discrepancy index that is considered a conservative metric (Ulrich & Gotelli, 2007) using the web-based program *NeD* (available at <http://ecosoft.alwaysdata.net/>; Strona, Galli, Seveso, Montano, & Fattorini, 2014). The significance of the nestedness score was evaluated by comparing the discrepancy index to a distribution of scores based upon 500 randomized runs.

In summer, richness in intermittent and perennial reaches overlapped (intermittent taxa richness range: 9–27; permanent taxa richness range: 9–42), and therefore, we did not conduct a nested analysis for this sampling period. Instead, we compared assemblage composition between intermittent and perennial reaches using a multiresponse permutation procedure (MRPP). MRPP is a non-parametric test of the heterogeneity of predefined groups within multidimensional space. A significant test indicates that assemblages within groups are more similar to each other than predicted by chance. MRPP tests were conducted in PC-Ord, v. 7.02 (McCune & Mefford, 2015).

## 2.6 | Network diversity

At the network scale, taxa accumulation curves were generated to examine the increase in taxa richness with increasing number of reaches using the R package “vegan” and the *specaccum* function (Oksanen et al., 2016). Comparisons between intermittent and perennial reaches of total richness and richness of all groups at the network scale were made using the *c2cv* function in the R package “rich” (Rossi, 2016). *c2cv* uses a randomization procedure to develop a distribution of richness differences between two groups and then calculates a probability (*p* value) that randomized differences will be

greater than the observed difference based upon 999 randomized runs. *T* tests, variation  $\beta$ -diversity, and network diversity were all analysed in the statistical program R, v. 3.4.0 (R Core Team, 2013). For all analyses,  $\alpha$  was set at 0.10.

## 2.7 | Classifying sensitive taxa

To understand whether groups of taxa were sensitive (or insensitive) to intermittence, we calculated that ratio of intermittent to perennial richness for all taxa as well as each group of interest. Ratios were calculated for average  $\alpha$ -diversity as well as network diversity. Groups with intermittent to perennial ratios similar to the ratio for all taxa were considered neither sensitive nor insensitive to intermittence. However, if the group's ratio was largely different from the ratio for all taxa, then that group was considered sensitive (if the group's ratio was smaller) or insensitive (if the group's ratio was larger) to intermittence. As a benchmark, we considered a difference of at least 0.20 in a group's ratio relative to the ratio for all taxa as an indicator that a group was either particularly sensitive or insensitive.

# 3 | RESULTS

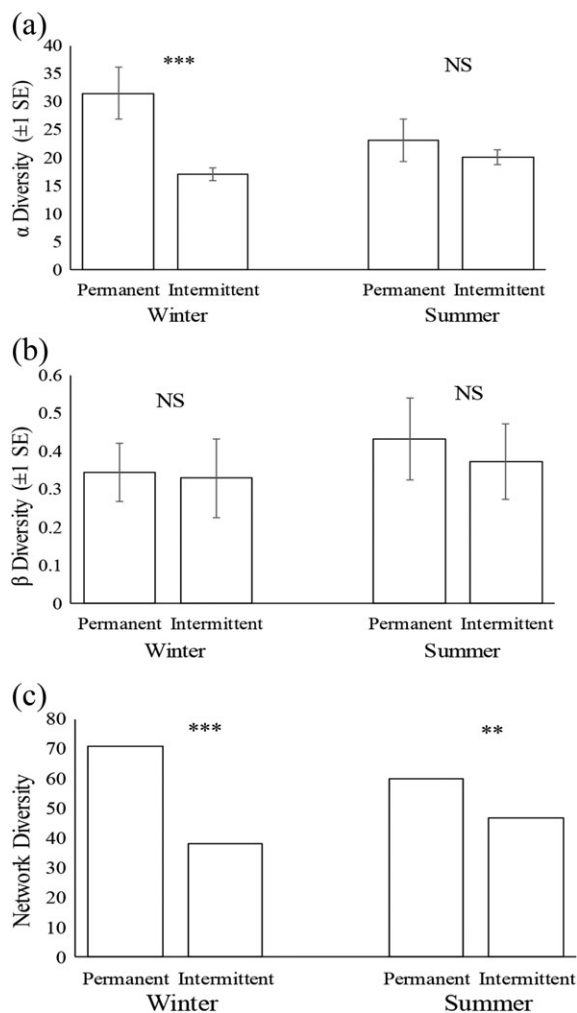
## 3.1 | $\alpha$ -Diversity

A total of 76 distinct taxa were identified across all samples in winter and 70 in summer. At the reach scale, permanent reaches had significantly more taxa than intermittent reaches in winter ( $t = 7.38$ ,  $p < 0.001$ ; Figure 2a). However, no significant difference existed in mean taxa richness between permanent and intermittent sites in summer ( $t = 0.76$ ,  $p = 0.47$ ; Figure 2a). Taxonomic richness of all six groups was significantly higher in permanent than intermittent reaches in winter ( $t > 3.80$  and  $p \leq 0.002$  in all cases; Figure 3a), but no significant differences were found for any of the groups in summer ( $t < 1.70$  and  $p > 0.17$  in all cases; Figure 4a).

At the reach scale, taxa richness in intermittent reaches was only 54% of that in permanent reaches in winter for all taxa combined (Table 1). Groups of macroinvertebrates had ratios that were similar to this with the exception of rare taxa. The richness of rare taxa in intermittent reaches was only 21% of that of all taxa (Table 1). Although not statistically significant, a similar pattern in the ratios of richness in intermittent and perennial reaches was observed in summer (Table 1). During summer, only the proportional richness of rare taxa in intermittent reaches (38%) was markedly lower than the proportional richness of all taxa (87%).

## 3.2 | $\beta$ -Diversity

No differences in variation  $\beta$ -diversity were found during either season for the entire macroinvertebrate assemblage (winter:  $F = 0.01$ ,  $p = 0.92$ ; summer:  $F = 0.59$ ,  $p = 0.45$ ; Figure 2b) or for any groups of taxa ( $F < 1.70$  and  $p > 0.20$  in all cases; Figures 3b and 4b). Variation  $\beta$ -diversity was not calculated for rare taxa during either season because PCoA solutions were not found for this group. Winter assemblages in intermittent reaches were different from those in perennial reaches (MRPP:  $A = 0.12$ ,  $p < 0.001$ ; Figure 5a). However, nestedness

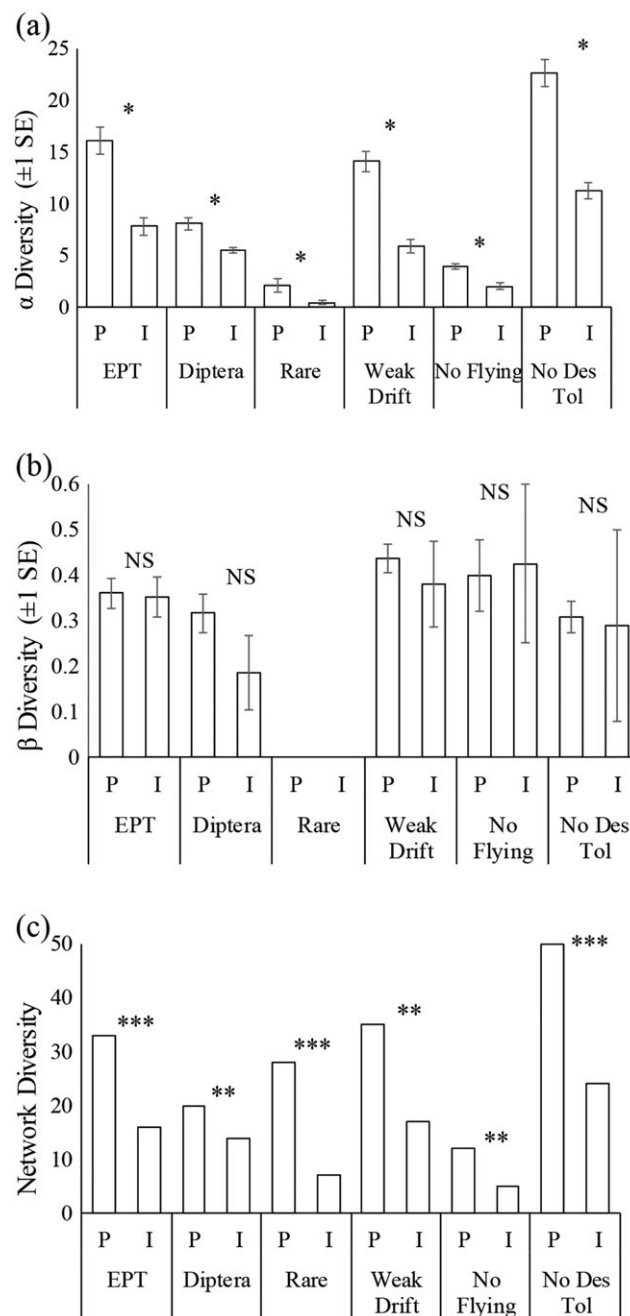


**FIGURE 2** Macroinvertebrate diversity in permanent and intermittent reaches during the two sampling seasons for (a)  $\alpha$ -diversity at the reach scale, (b) variation  $\beta$ -diversity (average distance to centroid), and (c) network diversity. NS: not significant

analysis revealed that assemblages in intermittent reaches were nested subsets of those found in relatively taxa-rich perennial reaches (Brualdi and Sanderson index = 110.0,  $z$  score = -22.6,  $p < 0.001$ ). Therefore, the separation of intermittent and perennial assemblages in the PCoA reflected the absence of taxa in intermittent reaches that were found in perennial reaches during winter. In summer, no difference in assemblage composition existed between assemblages in intermittent and perennial reaches (Figure 5b; MRPP:  $A < 0.001$ ,  $p = 0.36$ ).

### 3.3 | Network richness

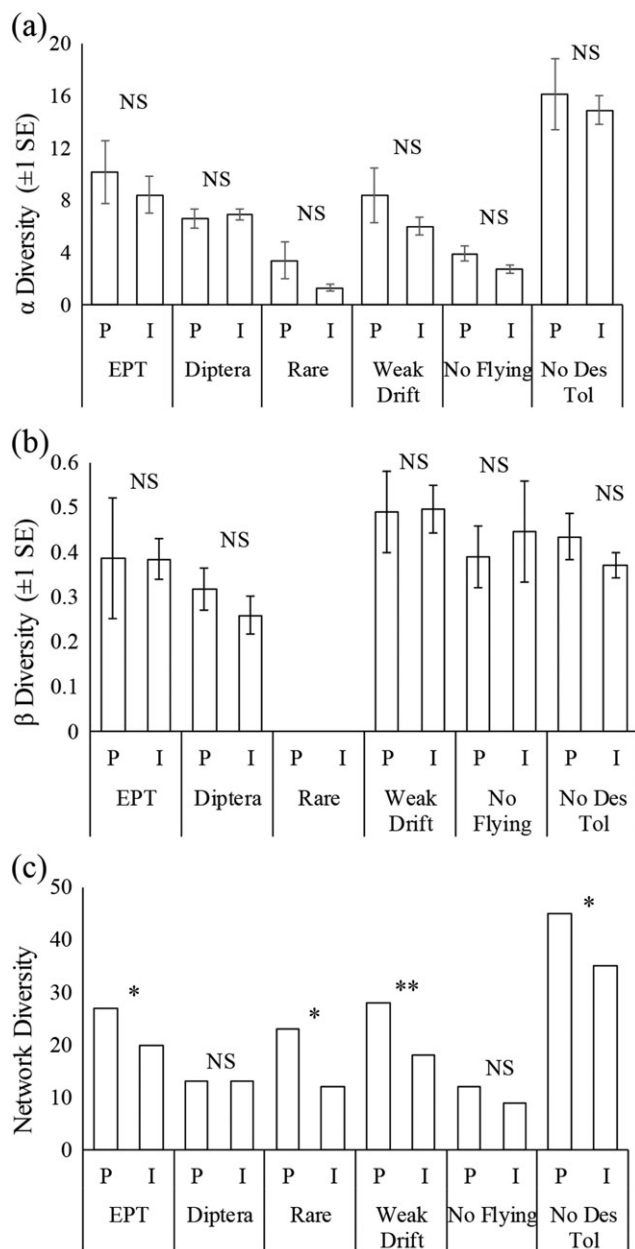
As the number of reaches increased, taxa richness accumulated in permanent reaches at a more rapid rate than in intermittent reaches during both summer and winter (Figure 6a,b). This pattern was evident for all groups as well, with the exception of Diptera in summer, when accumulation of richness was similar for both intermittent and permanent reaches (Figure 7). In winter, network richness for all taxa combined as well as for the six groups was significantly higher in the population of perennial reaches than in intermittent reaches



**FIGURE 3** Diversity of groups of macroinvertebrates comparing perennial and intermittent reaches during the winter season sampling; (a)  $\alpha$ -diversity at the reach scale, (b) variance  $\beta$ -diversity, and (c) network diversity. Because of the data structure for rare taxa, an ordination solution was not found for this group, and therefore, variance  $\beta$ -diversity was not calculated. Statistical comparisons were made between perennial and intermittent reaches for each group separately. “\*” indicates a  $p$  value of  $\leq 0.10$ , “\*\*\*” indicates a  $p$  value of  $\leq 0.05$ , “\*\*\*\*” indicates a  $p$  value of  $\leq 0.01$ , and “NS” stands for “not significant.” “EPT” stands for Ephemeroptera, Plecoptera, and Trichoptera taxa, “No Flying” are invertebrates with no flying stage, and “No Des Tol” stands for “no desiccation tolerance”

(Table 2). This pattern was also observed in the summer, except for Diptera, for which richness did not significantly differ (Table 2).

At the network scale, total taxa richness in intermittent reaches was 53% of that in permanent reaches in winter (Table 1). The most



**FIGURE 4** Diversity of groups of macroinvertebrates comparing perennial and intermittent reaches during the summer season sampling; (a)  $\alpha$ -diversity at the reach scale, (b) variance  $\beta$ -diversity, and (c) network diversity. Because of the data structure for rare taxa, an ordination solution was not found for this group, and therefore, variance  $\beta$ -diversity was not calculated. See Figure 3 for key to labels and significance levels

sensitive group to intermittency was rare taxa as intermittent reaches had only 25% of the taxa richness found in permanent reaches (Table 1). In contrast, dipterans were the least sensitive as richness of this group in intermittent reaches was 70% of that in permanent reaches (Table 1). Similar results were found in summer, when total richness in intermittent reaches was 78% of that in permanent reaches (Table 1). Rare taxa were again the most sensitive group, with a richness in intermittent reaches being 52% of that found in permanent reaches. In addition, Diptera were the least sensitive, with richness being the same in intermittent and permanent reaches during the summer (Table 1).

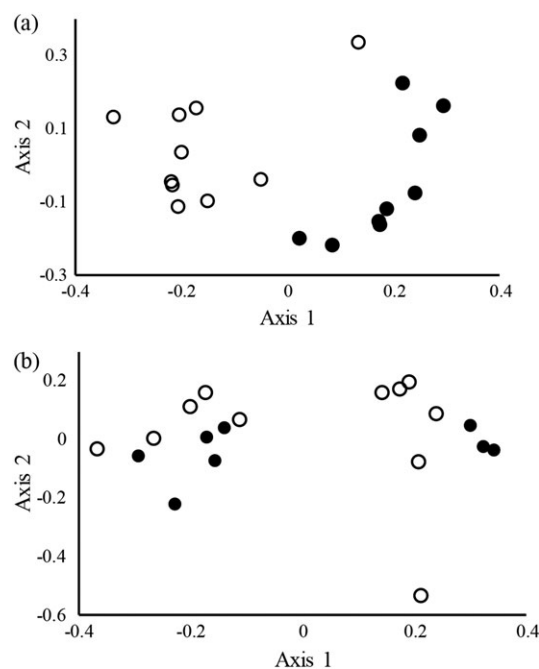
**TABLE 1** The ratio of intermittent taxa richness to perennial taxa richness for the entire assemblage ("Total") and for the different groups examined

|                 | Total | EPT  | Diptera | Rare | Weak drift | No flying | No des tolerance |
|-----------------|-------|------|---------|------|------------|-----------|------------------|
| Winter $\alpha$ | 0.54  | 0.48 | 0.68    | 0.21 | 0.42       | 0.51      | 0.50             |
| Summer $\alpha$ | 0.87  | 0.82 | 1.00    | 0.38 | 0.71       | 0.69      | 0.92             |
| Winter Network  | 0.53  | 0.48 | 0.70    | 0.25 | 0.49       | 0.42      | 0.48             |
| Summer Network  | 0.78  | 0.74 | 1.00    | 0.52 | 0.64       | 0.75      | 0.78             |

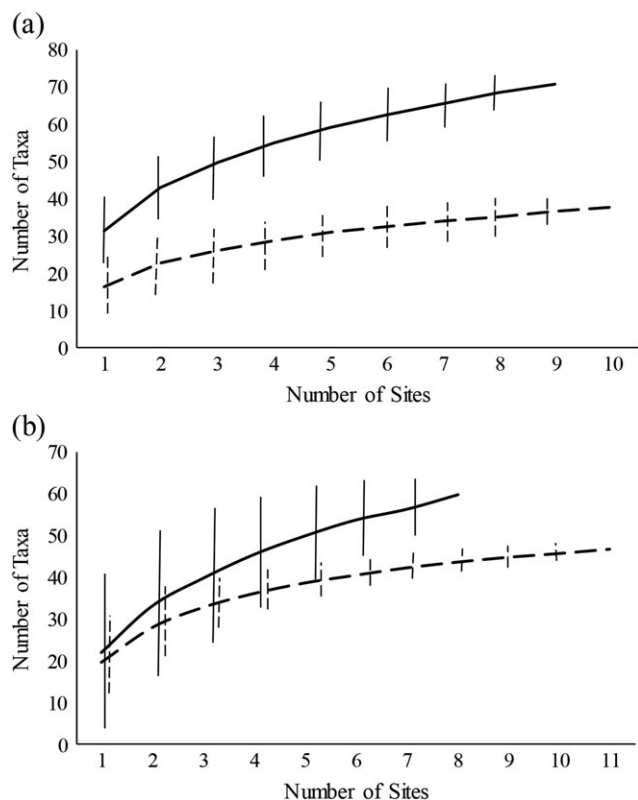
Note. Ratios for both  $\alpha$ -diversity and network diversity are shown for each of the two sampling seasons. "No flying" indicates macroinvertebrate taxa with no flying stage, and "No des tolerance" indicates taxa with no desiccation tolerance. EPT: Ephemeroptera, Plecoptera, and Trichoptera.

## 4 | DISCUSSION

In the Mission Creek system,  $\alpha$ -diversity was lower in intermittent reaches than nearby perennial reaches 5–6 weeks after the resumption of flow. However, after intermittent reaches had been flowing for ~6 months,  $\alpha$ -diversity in intermittent reaches recovered in the sense that this diversity was statistically indistinguishable from that found in nearby perennial reaches. Recovery of macroinvertebrate diversity has been observed in other intermittent reaches after the resumption of flow and can take as little as 1 week (Fowler, 2004) to several months (de Szoek et al., 2016). We defined recovery as a lack of statistical difference in  $\alpha$ -diversity between intermittent and perennial reaches, which is a common means for interpreting a lack of difference in diversity between intermittent and perennial reaches (Bogan et al., 2013; Bonada, Rieradevall, & Prat, 2007; Datry et al.,



**FIGURE 5** Principle coordinate analyses showing sites in taxa space for (a) winter and (b) summer. Filled circles indicate perennial sites, and open circles indicate intermittent sites



**FIGURE 6** Taxa accumulation curves for (a) winter and (b) summer sampling. Perennial reaches are indicated with solid lines and intermittent reaches with dashed lines. Vertical lines encompass 95% confidence intervals about the means

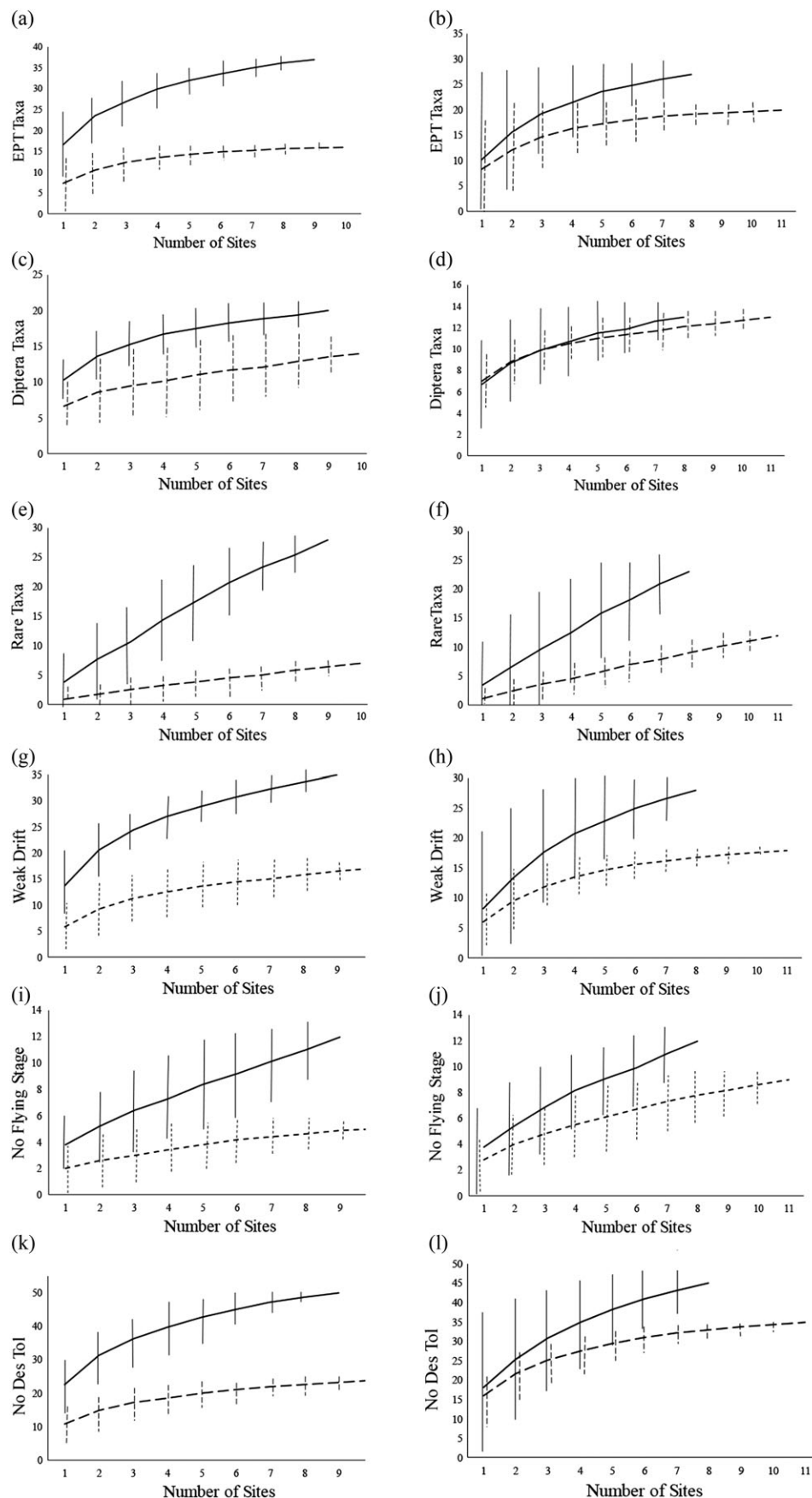
2016). However, variability in diversity among sites can lead to a lack of statistical differences and potentially mask patterns of diversity at other spatial scales.

In contrast to  $\alpha$ -diversity, we predicted that  $\beta$ -diversity would be higher in intermittent reaches during winter because of the brief amount of time for mixing of taxa among sites (Larned et al., 2010; Rolls et al., 2016; Schriever & Lytle, 2016; Stubbington et al., 2017). However, we predicted that after flowing for 6 months,  $\beta$ -diversity would be similar between intermittent and perennial reaches as taxa would have had many opportunities to mix among all reaches. In contrast to our predictions, no differences existed in  $\beta$ -diversity during either season. It is not clear why  $\beta$ -diversity was similar between the two types of reaches during winter. However, both dispersal dynamics and environmental similarity among the intermittent reaches likely played a role. All intermittent sites in our study were in close proximity to permanent reaches (500–1,100 m) and were dry for similar durations (~175–200 days), allowing for similar rates of colonization and mixing among both intermittent and permanent sites after flow resumption. In addition, all sites drained through similar geology and land use, further minimizing the amount of environmental heterogeneity among sites. Although this lack of difference in  $\beta$ -diversity during winter was not predicted, a recent meta-analysis using data from Europe and North America also found no difference in  $\beta$ -diversity between intermittent and permanent reaches (Rosset, Ruhi, Bogan, & Datry, 2017). Thus, a relatively high  $\beta$ -diversity in intermittent reaches might not be a common pattern as predicted by conceptual models (Larned et al., 2010; Stubbington et al., 2017).

Even though intermittent reaches had low  $\alpha$ -diversity during winter, it is possible that these reaches harbour specialist taxa not found in perennial reaches and thus contribute to larger scale diversity. Intermittent reaches of headwater stream systems in western Oregon and Arizona have been found to contain taxa with traits (e.g., egg and larval diapause) that allow an exploitation of intermittent reaches (Bogan et al., 2013; Dieterich & Anderson, 2000). However, in other systems, taxa-poor assemblages in intermittent reaches are nested subsets of the assemblages found in perennial reaches (Arscott et al., 2010; Datry, 2012; Datry et al., 2014). This pattern was found in the Mission Creek system in winter. Common, generalist taxa (e.g., chironomid larvae, Simuliidae larvae, Baetis nymphs, and Blephariceridae larvae) found throughout all stream reaches dominated intermittent reaches. Although this pattern might indicate that intermittence is a filter permanently excluding taxa from intermittent reaches, it is more likely that the nested pattern is indicative of a point in the recovery of assemblages in intermittent reaches. Indeed, in summer, the assemblages in intermittent reaches recovered to that of nearby perennial reaches given the similarity between the two types of reaches as measured by  $\alpha$ -diversity and assemblage composition.

During winter, network diversity was lower in intermittent reaches than permanent reaches. This result was expected given the low  $\alpha$ -diversity and nestedness of intermittent reaches. However, during summer, a different pattern emerged. Despite similarities in both  $\alpha$ - and  $\beta$ -diversity, network diversity continued to be lower in intermittent reaches than perennial reaches (60 taxa in the population of perennial reaches and 47 in the intermittent reaches). Given the similarities in  $\alpha$ -diversity between intermittent and perennial reaches, we would expect  $\beta$ -diversity to be higher in perennial reaches if network diversity is higher. The metric of  $\beta$ -diversity used here—variance  $\beta$ -diversity—measured the variability in assemblage composition among sites by taking into account both taxa gains and losses as well as the sharing of taxa. In contrast, as illustrated by the species area curves, network diversity considers only the gains of new taxa across sites and reveals that unexpected patterns in diversity can emerge at this scale. Few studies have compared network diversity between intermittent and permanent reaches and thus whether the patterns observed in the Mission Creek system are common is not clear. In contrast to our results, Storey et al. (2011) found that although intermittent and permanent reaches of New Zealand streams had similar  $\alpha$ -diversities, intermittent reaches had both higher  $\beta$  and network diversities. The authors attributed higher  $\beta$  and network diversities to greater habitat diversity in intermittent reaches.

The taxa lost from intermittent reaches are not, necessarily, a random subset of the regional taxa pool (Arscott et al., 2010). Some taxa are relatively sensitive to intermittency, whereas others are resistant or resilient. Mayflies, stoneflies, and caddisflies are, in general, large-bodied taxa sensitive to a variety of stressors (Wallace, Grubaugh, & Whiles, 1996), including intermittency (Arscott et al., 2010; Datry et al., 2014). In the Mission Creek system, the richness of EPT taxa was lower in intermittent reaches than permanent reaches at the reach scale (during winter only) and at the network scale (during summer and winter). However, the proportional decline of EPT taxa in intermittent reaches was similar to the proportional



**FIGURE 7** Taxa accumulation curves for winter (left-hand column) and summer (right-hand column). Solid lines indicate perennial reaches and dashed lines intermittent. Vertical lines encompass 95% confidence intervals. Panels (a) and (b) Ephemeroptera, Plecoptera, and Trichoptera (EPT) taxa, (c) and (d) Diptera taxa, (e) and (f) rare taxa, (g) and (h) taxa with weak drift propensity, (i) and (j) taxa with no flying stage, and (k) and (l) taxa with no desiccation tolerance

**TABLE 2** Richness at the landscape scale for permanent and intermittent reaches for all taxa ("Total") and six groups

| Metric                   | Permanent richness | Intermittent richness | Observed difference | Predicted difference | p value |
|--------------------------|--------------------|-----------------------|---------------------|----------------------|---------|
| Winter                   |                    |                       |                     |                      |         |
| Total                    | 71                 | 38                    | 32                  | -2.2                 | 0.001   |
| EPT                      | 33                 | 16                    | 17                  | -0.6                 | 0.011   |
| Diptera                  | 20                 | 14                    | 6                   | -0.6                 | 0.029   |
| Weak drift               | 35                 | 17                    | 18                  | -1.0                 | 0.001   |
| No flying stage          | 12                 | 5                     | 7                   | -1.0                 | 0.036   |
| No desiccation tolerance | 50                 | 24                    | 26                  | -1.5                 | 0.001   |
| Rare                     | 28                 | 7                     | 21                  | -1.6                 | 0.004   |
| Summer                   |                    |                       |                     |                      |         |
| Total                    | 60                 | 47                    | 13                  | -6.1                 | 0.043   |
| EPT                      | 27                 | 20                    | 7                   | -1.8                 | 0.051   |
| Diptera                  | 13                 | 13                    | 0                   | -1.2                 | >0.50   |
| Weak drift               | 28                 | 18                    | 10                  | -2.9                 | 0.022   |
| No flying stage          | 12                 | 9                     | 3                   | -2.0                 | 0.10    |
| No desiccation tolerance | 45                 | 35                    | 10                  | -3.8                 | 0.083   |
| Rare                     | 23                 | 12                    | 11                  | -4.8                 | 0.085   |

Note. Predicted differences are based on the "Rich" analyses using Monte Carlo simulations (999 randomized runs), and p values were generated from the simulations. EPT: Ephemeroptera, Plecoptera, and Trichoptera.

decline of all taxa, indicating that this group was not particularly sensitive to intermittency.

In contrast to EPT taxa, many dipterans are tolerant taxa whose abundance and relative taxa richness increase in intermittent reaches in other systems (Bogan et al., 2013; Datry et al., 2014). In our study, dipteran richness decreased during winter at both the reach and network scale. However, the proportional decline in dipteran richness was less than that found for all taxa, indicating that these taxa are less sensitive to intermittency than the entire assemblage. In summer at the landscape scale, dipteran richness was the same in both intermittent and permanent reaches indicating recovery of this group. It is not clear why many dipterans are relatively insensitive to intermittency; however, many dipteran taxa possess traits conferring rapid dispersal (Pilière et al., 2016; Rader, 1997). In addition, many dipterans are abundant taxa with high fecundity (Courtney & Merritt, 2008; Huryn, Anderson, & Wallace, 2008) and thus produce a large number of propagules. This combination of high dispersal rates and high abundance should confer great resilience to intermittency. However, not all dipterans were abundant in our system. As discussed below, rare taxa were sensitive to intermittence and dipterans made up 24% of the rare taxa during both winter and summer.

Traits conferring either resistance or resilience to intermittency are considered primary determinants of the recolonization of intermittent reaches by macroinvertebrates (Bogan, Hwan, Cervantes-Yoshida, Ponce, & Carlson, 2017; Robson et al., 2011). Therefore, we predicted that macroinvertebrates lacking either strong resilience (e.g., poor dispersal) or resistance (e.g., no dessication tolerance) traits would be particularly sensitive to flow intermittency. However, in the Mission Creek system, the richness of poor dispersers and taxa lacking dessication tolerance did not decline to a greater degree than the entire assemblage, indicating that these groups were not particularly sensitive to flow intermittency. Although this lack of an effect was somewhat surprising, specific traits such as dessication tolerance and

possession of a flying stage are components of life history strategies that involve suites of traits. These suites of traits can determine whether a taxon is sensitive or not to a specific stressor (Verberk, van Noordwijk, & Hildrew, 2013), including flow intermittency.

In contrast to the trait groups, rare taxa appeared particularly sensitive to flow intermittence as the proportional loss of rare taxa in intermittent reaches was greater than overall taxa loss. This pattern was observed during both seasons and at both the reach and landscape scale (although not significant at the reach scale during summer, intermittent reaches had on average 67% fewer taxa during this season than permanent reaches). Although the sensitivity of rare taxa might stem from shared taxonomy or sets of traits related to resilience or resistance, there was little evidence that a specific taxonomic group or small set of groups dominated rare taxa. During both seasons, rare taxa were found in over 50% of all families and orders identified (see Table A1). In addition, trait composition was similar between rare taxa and the entire assemblage indicating that rare taxa were not, in general, more likely to have poor dispersal traits or lack dessication tolerance than the entire assemblage (Table A1). Thus, rarity per se appears to be an important driver of vulnerability to intermittency. Metapopulation models indicate that the colonization rate of unoccupied patches is positively related to commonness of a taxon (i.e., the number of occupied patches; Gotelli & Taylor, 1999). The small number of reaches occupied by rare taxa combined with their low densities indicates that these taxa probably produce low numbers of propagules. Therefore, they are less likely to colonize intermittent reaches than common taxa regardless of their dispersal abilities. We are not aware of research on macroinvertebrates that has examined the relationship between abundance, dispersal ability, and colonization of intermittent or disturbed reaches. However, Albanese, Angermeier, and Peterson (2009) found that abundance plays a more important role than dispersal ability in driving the colonization rates of unoccupied reaches by stream fish.

In conclusion, in the Mission Creek system, the macroinvertebrate assemblage of intermittent reaches did not recover to that of nearby permanent reaches despite similarities in  $\alpha$ - and  $\beta$ -diversity between the two types of reaches. Lack of recovery was evident only when taking a relatively large, stream network view and was masked by between reach variability at smaller spatial scales. Of the different macroinvertebrate groups considered, only rare taxa displayed a consistent sensitivity to intermittence in the sense that richness of this group was particularly low in intermittent reaches compared with overall richness and that of other groups. In the Mission Creek system, a large proportion of the taxa were considered rare (43% in winter and 47% in summer). This pattern of rare taxa comprising a high proportion of total richness is common in many stream systems (Cao, Williams, & Williams, 1998; Resh, Beche, & McElravy, 2005). The spatial and temporal extent of intermittent reaches is predicted to increase in many watersheds in the future with climate change and increasing demands on surface water (Larned et al., 2010). Our results indicate that understanding the dynamics of intermittent reaches and the potential loss of biodiversity with increasing intermittency will be improved through a consideration of multiple spatial scales, taxa abundance/rarity, dispersal abilities, and recovery dynamics through the wet season.

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## ORCID

David E. Wooster  <http://orcid.org/0000-0001-8028-3909>

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## APPENDIX A.

**TABLE A1** Characteristics of rare taxa and all taxa combined for winter and summer

| Characteristic                                | Winter     |             | Summer    |            |
|-----------------------------------------------|------------|-------------|-----------|------------|
|                                               | Rare       | All taxa    | Rare      | All taxa   |
| EPT taxa                                      | 39%        | 45%         | 27%       | 40%        |
| Diptera                                       | 24%        | 29%         | 24%       | 24%        |
| Weak drift                                    | 42%        | 47%         | 51%       | 44%        |
| No flying stage                               | 30%        | 18%         | 30%       | 23%        |
| No dessication tolerance                      | 81%        | 76%         | 73%       | 78%        |
| Number of families                            | 26         | 51          | 28        | 49         |
| Number of orders                              | 9          | 13          | 11        | 16         |
| Density as #/m <sup>2</sup> (SE) <sup>a</sup> | 22.4 (9.1) | 55.9 (12.5) | 5.1 (1.1) | 27.1 (9.9) |

Note. EPT: Ephemeroptera, Plecoptera, and Trichoptera.

<sup>a</sup>Average density was calculated only for sites in which taxa occurred (i.e., no 0 values).